

Multiple Choice

1. **Answer:** B

Options: A) $P(E, F), P(H), P(E|H), P(F|H)$; B) $P(E, F), P(H), P(E, F|H)$; C) $P(H), P(E|H), P(F|H)$; D) $P(E, F), P(E|H), P(F|H)$

Note: The correct answer is based on the application of Bayes' theorem, which requires the joint probability $P(E, F)$, the prior probability $P(H)$, and the conditional probability $P(E, F|H)$ to compute the posterior probability $P(H|E, F)$ without any assumptions of independence.

2. **Answer:** A

Options: A) Linear hard-margin SVM.; B) Linear Logistic Regression.; C) Linear Soft margin SVM.; D) The centroid method.

Note: The linear hard-margin SVM is designed specifically to find a hyperplane that perfectly separates linearly separable data without any misclassifications, making it applicable only in scenarios where such separation is achievable.

3. **Answer:** A

Options: A) Supervised learning; B) Unsupervised learning; C) Clustering; D) None of the above

Note: The correct answer is "supervised learning" because predicting rainfall involves using labeled historical data to train a model that can make accurate forecasts based on observed features or cues.

4. **Answer:** A

Options: A) linear in K ; B) quadratic in K ; C) cubic in K ; D) exponential in K

Note: K -fold cross-validation is considered linear in K because the model is trained and validated K times, resulting in a computational complexity that increases linearly with the number of folds.

5. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: The statement is false because regression analysis identifies relationships between variables and predicts outcomes but does not establish causation, as correlation does not imply causation.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: DCGANs primarily utilize convolutional layers and batch normalization rather

than self-attention mechanisms to improve image generation quality and training stability.

7. **Answer:** False

Options: A) True; B) False

Note: The classification run time of the nearest neighbors algorithm is $O(N)$ in the worst case because it requires examining all N training data points to find the nearest neighbor, not $O(\log N)$ as suggested by the statement.

8. **Answer:** True

Options: A) True; B) False

Note: The use of weak classifiers in bagging helps prevent overfitting because averaging their predictions reduces variance and leads to a more robust model that generalizes better to unseen data.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because a Bayesian network with that structure requires specifying the conditional probabilities for each node, which results in 8 independent parameters due to the dependencies and independence constraints defined by the network's structure.

10. **Answer:** True

Options: A) True; B) False

Note: ImageNet includes millions of images collected from the internet, featuring a wide range of resolutions and aspects, which contributes to its diversity and effectiveness as a dataset for training deep learning models.

Long Form Response

11. **Answer:** `def swap_count(s): | return swap`

Note: The answer correctly identifies that the function 'swap_count(s)' is designed to compute the minimum number of swaps needed to balance brackets in the given string, which is a critical task in algorithms related to string manipulation and balance checking.

12. **Answer:** `def nth_items(list,n): | return list[:n]`

Note: correct because the function uses Python's slicing feature to generate a new list that contains every n th element from the original list, efficiently fulfilling the requirement to select specific items based on their position.

End of Answer Key